

⇒ All the theorems in ar-10 book.

① Pythagoras Theorem

$$= a^2 + b^2 = c^2$$

- The Square of the hypotenuse equals the Sum of the Squares of the other two sides

② Parallel Line Angle Theorems.

Transversal

~~two~~ Cut two parallel lines

- Alternate angles are equal
- Corresponding angles are equal
- Co-interior angles sum to 180°

③ Isosceles Triangle theorem

- If two sides of a triangle are equal, then the angles opposite those sides are equal.

$$\text{eg: } AB = AC \\ \hat{B} = \hat{C}$$

④ Angle Sum of a triangle

- The Sum of the interior angles of a triangle is 180°

$$A + B + C = 180^\circ$$

⑤ Exterior Angle theorem

- An exterior angle of a triangle equals the Sum of the two opposite interior angles

⇒ Congruent Triangle Theroms

⑥ SSS (Side-Side-Side)

- If three sides of ^{one} triangle equal three sides of another triangle then they are congruent.

⑦ SAS (Side-Angle-Side)

- If two sides and the included angle are equal then they are congruent.

⑧ ASA (Angle-Side-Angle)

- If two angles and the included side are equal then the triangles are congruent.

⑨ RHS (Right angle-hypotenuse-Side)

- For a right angled triangle
 - one angle = 90°
 - hypotenuse equal
 - one side equal

∴ triangles are congruent